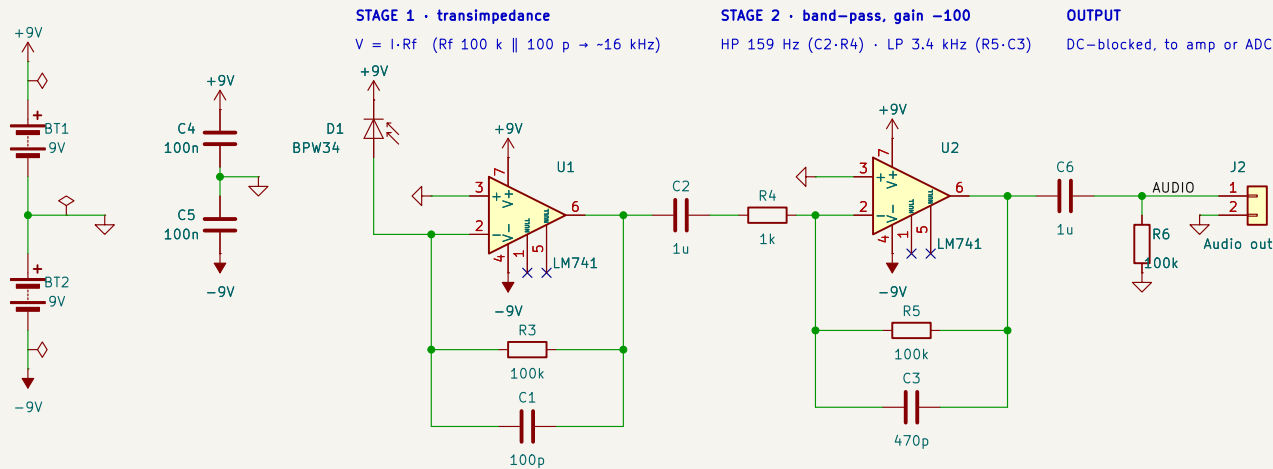


Laser microphone receiver – design study (unbuilt)

A Class 2 laser ($\leq 1\text{ mW}$, 650 nm) reflects off a window pane; sound moves the glass by nanometres, moving the reflected spot across the photodiode. The receiver turns that light change into audio.



Parts chosen from the Pegboard inventory where possible: U1-U2 (2× LM741), R3/R5/R6 (100k), R4 (1k), C2/C6 (1 μ F).
To source: D1 BPW34, C1 100 pF, C3 470 pF, C4-C5 100 nF, laser module, battery clips. LM741 is noisy and slow;
a TL072 is the planned upgrade. Never aim the laser at people or at windows you do not own.

Not yet built, measured or SPICE-verified against hardware
Photodiode TIA + band-pass, $\pm 9\text{ V}$

Nicholas Jordan

Sheet: /
File: laser-mic.kicad_sch

Title: Laser microphone receiver

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.6

Rev: A – design study, unbuilt

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